

Measurement-Based Extraction and Statistical Modeling of GPS L1 C/A Multipath Components in Urban and Mountain/Valley Environments

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Abstract—Multipath propagation in a time-varying satellite-to-ground channel produces delayed and Doppler-shifted replicas of the direct GPS signal. This paper presents a measurement-based method to extract individual multipath components (MPCs) and characterize their parameters in Urban and Mountain/Valley environments. Navigation data form coherent 40-ms observations for each tracked satellite. The space-alternating generalized expectation-maximization (SAGE) algorithm estimates the delay, Doppler, and complex gain of candidate MPCs. Temporal consistency selects persistent MPC observations for characterization. The analysis uses 518 persistent MPC observations from 236 satellite tracks. Excess delay is described by environment-specific lognormal distributions, signed relative Doppler by a track-balanced empirical cumulative distribution, and relative power by a two-component Gaussian mixture model. A connected two-line model describes the mean decrease in relative power with excess delay. These descriptions provide measurement-based inputs for simplified GNSS channel simulations and receiver evaluation.

Index Terms—GNSS multipath, GPS L1 C/A, SAGE, channel modeling, delay, Doppler.

I. INTRODUCTION

Global Navigation Satellite Systems (GNSS) support vehicle navigation and intelligent transportation. A moving receiver observes a time-varying satellite-to-ground channel. Reflections from buildings, terrain, and vehicles produce signal replicas with different delays, Doppler shifts, and powers. These parameters vary as the receiver moves [1], [2], [3].

Earlier GPS multipath studies have characterized receiver-level effects through carrier-to-noise-density ratio, pseudorange error, and carrier-phase error [2], [4], [5]. These quantities indicate changes caused by multipath, but they do not provide the delay, Doppler, and gain of individual MPCs. Path-level parameters are useful for statistical channel modeling and controlled receiver evaluation.

GNSS multipath models address different levels of the problem. Jost et al. modeled indoor pseudorange errors with a two-component Gaussian mixture, and Khider et al. applied this error model to multisensor positioning [6], [7]. These receiver-oriented models describe ranging errors rather than individual propagation paths. Lehner and Steingass used measured reflection delay, power, duration, and Doppler to construct a time-varying land-mobile satellite channel [8]. At the path-parameter level, SAGE estimates individual MPC parameters from complex baseband measurements by updating subsets of channel parameters in sequence [9], [10]. Gaussian mixtures

can compactly describe a measured power distribution with more than one mode [11].

This work combines per-satellite path extraction and environment-specific statistical descriptions. Navigation-data alignment forms coherent GPS L1 C/A observations for each tracked satellite. SAGE estimates MPC delay, Doppler, and complex gain, and temporal consistency selects persistent MPC observations. The measured excess delay, signed relative Doppler, relative power, and power-delay relation are then described for Urban and Mountain/Valley. These descriptions provide measurement-based MPC parameters for simplified GNSS simulations. Section II describes the measurements and per-satellite processing, Section III presents MPC estimation and statistical descriptions, and Section IV presents the results.

II. MEASUREMENTS AND SIGNAL PROCESSING

A. Measurement Setup

The measurement system uses a TEST-TREE RF-Catcher V2 and a roof-mounted GNSS dome antenna. The antenna is right-hand circularly polarized and has a 40-dB active gain. The receiver records GPS L1 C/A at 1575.42 MHz [12]. The sampling rate is 10.23 MHz. Samples are stored as interleaved, little-endian, signed 16-bit I/Q values.

The measurements were collected while the vehicle traveled along two types of roads. Urban routes are surrounded by dense buildings, whereas Mountain/Valley routes pass through mountainous or valley terrain.

B. Per-Satellite Multipath Processing

GNSS-SDR first tracks each visible satellite and decodes its navigation data [13]. Because different satellites use different pseudorandom-noise codes, each tracked satellite is processed in a separate signal stream. Fig. 1 summarizes this processing flow. Each GPS L1 C/A navigation bit lasts 20 ms; thus, a 40-ms observation may contain a bit transition that reverses the signal sign. The decoded bit signs are removed before coherent observations are formed. Correlation screening identifies observations that may contain an additional signal component. SAGE then estimates MPC parameters for each selected observation. A temporal-consistency check compares adjacent observations and retains persistent MPC observations for the statistical analysis.

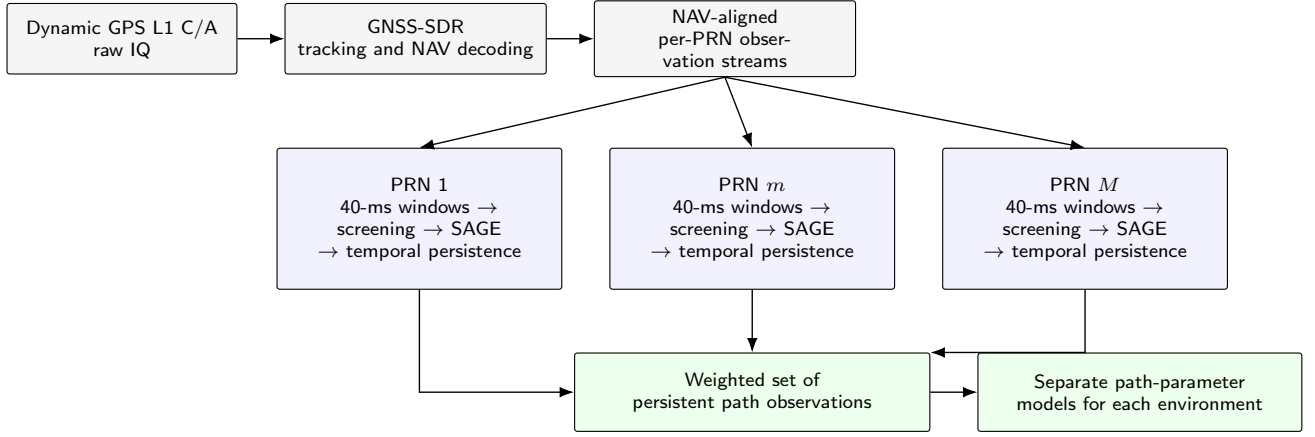


Fig. 1. Measurement and per-satellite MPC processing flow.

III. MULTIPATH COMPONENT ESTIMATION AND STATISTICAL MODELING

A. SAGE-Based MPC Estimation

Let m denote one tracked satellite-to-receiver channel. Its complex baseband signal is

$$r_m(t) = \sum_{\ell=0}^{L_m-1} \alpha_{m,\ell} s_m(t - \tau_{m,\ell}) e^{j2\pi\nu_{m,\ell}t} + n_m(t), \quad (1)$$

where $s_m(t)$ is the aligned C/A-code waveform and $n_m(t)$ is noise. MPC ℓ has complex gain $\alpha_{m,\ell}$, delay $\tau_{m,\ell}$, and Doppler frequency shift $\nu_{m,\ell}$. The earliest estimated component ($\ell = 0$) is the relative reference; it is not assumed to be a physically verified line-of-sight path.

The SAGE algorithm updates one MPC at a time [9], [10]. In intuitive terms, the current reconstructions of the other MPCs are subtracted before the target MPC is re-estimated. For target MPC ℓ , the E-step forms a conditional hidden-data estimate by subtracting the current contributions of the other MPCs from the received observation:

$$\mathbf{r}_{m,\ell} = \mathbf{r}_m - \sum_{k \neq \ell} \hat{\alpha}_{m,k} \mathbf{q}_m(\hat{\tau}_{m,k}, \hat{\nu}_{m,k}). \quad (2)$$

The M-step updates the delay and Doppler frequency shift of the target MPC by

$$(\hat{\tau}_{m,\ell}, \hat{\nu}_{m,\ell}) = \arg \max_{\tau, \nu} \frac{|\mathbf{q}_m(\tau, \nu)^H \mathbf{r}_{m,\ell}|^2}{\mathbf{q}_m(\tau, \nu)^H \mathbf{q}_m(\tau, \nu)}, \quad (3)$$

where $\mathbf{q}_m(\tau, \nu)$ is a sampled code replica with fractional delay and Doppler. After all MPCs have been updated in one sweep, their complex gains are jointly estimated by least squares.

Equation (3) selects the delay–Doppler pair with the highest normalized correlation to the conditional hidden-data estimate. A coarse correlation search provides the initial parameter values. The delay is then refined with fractional-sample code replicas spaced by 0.1 sample within a ± 0.8 -sample neighborhood. The Doppler frequency shift is searched within ± 30 Hz of the tracking estimate in 5-Hz steps. The iterations terminate

after 10 sweeps or when the relative change in the residual sum of squares (RSS) falls below 10^{-6} .

For each 40-ms observation, four candidate models are evaluated, with one, two, three, or four MPCs. The selected path count is chosen by comparing the Bayesian information criterion and residual sum of squares (RSS) across these candidates. This comparison balances signal representation against model complexity.

B. Persistent MPC Observations

After local estimation, MPC estimates from adjacent observations are compared. A match is declared only when the excess-delay, signed relative Doppler, and relative-power differences are simultaneously no greater than 1.5 samples, 40 Hz, and 10 dB, respectively. An MPC is labeled persistent when matching estimates occur in at least three consecutive observations. A 40-ms observation is used for modeling only when every nonreference MPC satisfies this condition. The center-time parameters then form the persistent MPC observations.

For each persistent MPC observation, the three parameters are expressed relative to the reference component:

$$\Delta\tau = \tau_\ell - \tau_0, \quad \nu_{\text{rel}} = \nu_\ell - \nu_0, \quad P_{\text{rel}} = 10 \log_{10} \frac{|\alpha_\ell|^2}{|\alpha_0|^2}. \quad (4)$$

Here, $\Delta\tau$ is the delay relative to the reference component, ν_{rel} is signed relative Doppler, and P_{rel} is the power relative to the reference component in decibels.

C. Environment-Specific Statistical Descriptions

Satellite tracks contain different numbers of persistent MPC observations. Each track is therefore assigned the same total weight, preventing a longer track from dominating the environment-level statistics simply because it contains more observations. If track j contains n_j observations, each observation in that track receives weight $w_i = 1/n_j$.

Several parametric distributions were considered for excess delay and relative power in each environment. Candidate distributions were compared using leave-one-scene-out (LOSO)

validation: one complete measurement scene was held out, the candidates were fitted to the remaining scenes, and their performance was evaluated on the held-out scene. This was repeated until every scene had been held out once. Model selection used both the predictive likelihood and the maximum weighted CDF deviation,

$$D_w = \max_i \left| \hat{F}_w(x_i) - F_\theta(x_i) \right|, \quad (5)$$

where $\hat{F}_w$ is the weighted empirical CDF and F_θ is the model CDF. Here, D_w is used as a descriptive goodness-of-fit measure: a smaller value indicates closer agreement between the measured and modeled cumulative distributions.

Signed relative Doppler is represented directly by its track-balanced empirical CDF. For environment e , the CDF and its weighted quantiles are defined as

$$\begin{aligned} \hat{F}_{\nu,e}(x) &= \frac{\sum_{i \in e} w_i \mathbf{1}(\nu_i \leq x)}{\sum_{i \in e} w_i}, \\ P_q(e) &= \inf\{x : \hat{F}_{\nu,e}(x) \geq q\}. \end{aligned} \quad (6)$$

Here, ν_i is the signed relative Doppler of observation i . The empirical representation retains the measured signed relative Doppler values and is summarized by P_{10} , P_{50} , and P_{90} .

To represent the observed positive and right-skewed excess-delay distribution, a lognormal form is used:

$$\ln(\Delta\tau - \tau_s) \sim \mathcal{N}(\mu_\tau, \sigma_\tau^2), \quad (7)$$

where τ_s is a shift, $\exp(\mu_\tau)$ is the scale, and σ_τ controls the spread in the logarithmic domain. The Urban model uses an estimated shift, while the Mountain/Valley model uses $\tau_s = 0$.

The measured relative-power density contains two main modes. A two-component Gaussian mixture model (GMM) is used to represent these modal regions:

$$p(x) = \pi \mathcal{N}(x; \mu_1, \sigma_1^2) + (1 - \pi) \mathcal{N}(x; \mu_2, \sigma_2^2). \quad (8)$$

The mixing weight π gives the relative contribution of the first Gaussian component. The two Gaussian terms represent the two modal regions in the relative-power measurements.

Relative power is also modeled as a function of excess delay. Two line segments with different slopes meet at a common breakpoint τ_b :

$$P(\tau) = \begin{cases} P_b + b_1(\tau - \tau_b), & \tau < \tau_b, \\ P_b + b_2(\tau - \tau_b), & \tau \geq \tau_b. \end{cases} \quad (9)$$

This connected model allows the mean rate of power decrease to differ between low-delay and high-delay MPCs while keeping the mean trend continuous. The fitted delay and power distributions and the track-balanced empirical CDF of signed relative Doppler provide complementary parameter descriptions for each environment.

Receiver-level GPS multipath models describe quantities such as pseudorange error or carrier-to-noise-density ratio. Traditional land-mobile satellite channel models organize reflections in a time-varying channel impulse response or power-delay profile [8]. The present descriptions operate at the

path-parameter level: they provide MPC delay, signed relative Doppler, and power together with a mean power-delay relation. These measurement-based parameters can support simplified GNSS channel simulation and receiver evaluation.

The selected forms are also consistent with the main structures observed in the measured parameters: a lognormal density for positive delay with a long right tail, a two-component GMM for the two relative-power modes, and an empirical CDF for signed relative Doppler. The connected two-line model describes the mean power change with delay; its common breakpoint keeps the trend continuous while allowing different rates for shorter- and longer-delay MPCs.

IV. STATISTICAL MODELING RESULTS

A. Modeling Data and Excess Delay

The analysis uses 518 persistent MPC observations from 236 satellite tracks, 36 recordings, and 9 measurement scenes. Of these observations, 346 are from Urban and 172 are from Mountain/Valley.

Fig. 2(a) and (b) show that both excess-delay distributions are right-skewed. The Urban fit has a shift of 68.7 ns, a scale of 102.8 ns, and a log-domain standard deviation of 0.675. The Mountain/Valley fit has a scale of 187.8 ns and a log-domain standard deviation of 0.348. The larger Urban spread is associated with a broader right tail. Table I lists the model summaries and fit measures.

B. Signed Relative Doppler

Fig. 2(c) and (d) show the track-balanced empirical CDFs of signed relative Doppler. For Urban, the weighted P_{10} , P_{50} , and P_{90} values are -55.6 , 19.5 , and 80.6 Hz, respectively. The corresponding values for Mountain/Valley are -94.4 , -45.1 , and 55.3 Hz. The Mountain/Valley distribution is shifted toward more negative signed relative Doppler values, as indicated by its more negative P_{10} and median, whereas the Urban median remains positive. The empirical CDF preserves the measured concentration without imposing a parametric family.

C. Relative Power

Fig. 2(e) and (f) show two relative-power modes in both environments. These values are relative to the earliest component and are not absolute received powers. The means are near -13 dB for the weaker mode and -3 to -4 dB for the stronger mode. Compared with a single-Gaussian fit, the two-component GMM reduces D_w from 0.133 to 0.045 in Urban and from 0.137 to 0.037 in Mountain/Valley.

1) *Use in Simplified Channel Simulation:* For a selected environment, excess delay and relative power can be sampled from their fitted marginals, and signed relative Doppler can be resampled from its track-balanced empirical CDF. The connected power-delay model supplies a descriptive mean trend when delay dependence is needed. These values define an MPC in (1) and can be applied separately to each simulated satellite channel. The descriptions serve as marginal inputs to simplified simulations; they do not specify MPC occurrence

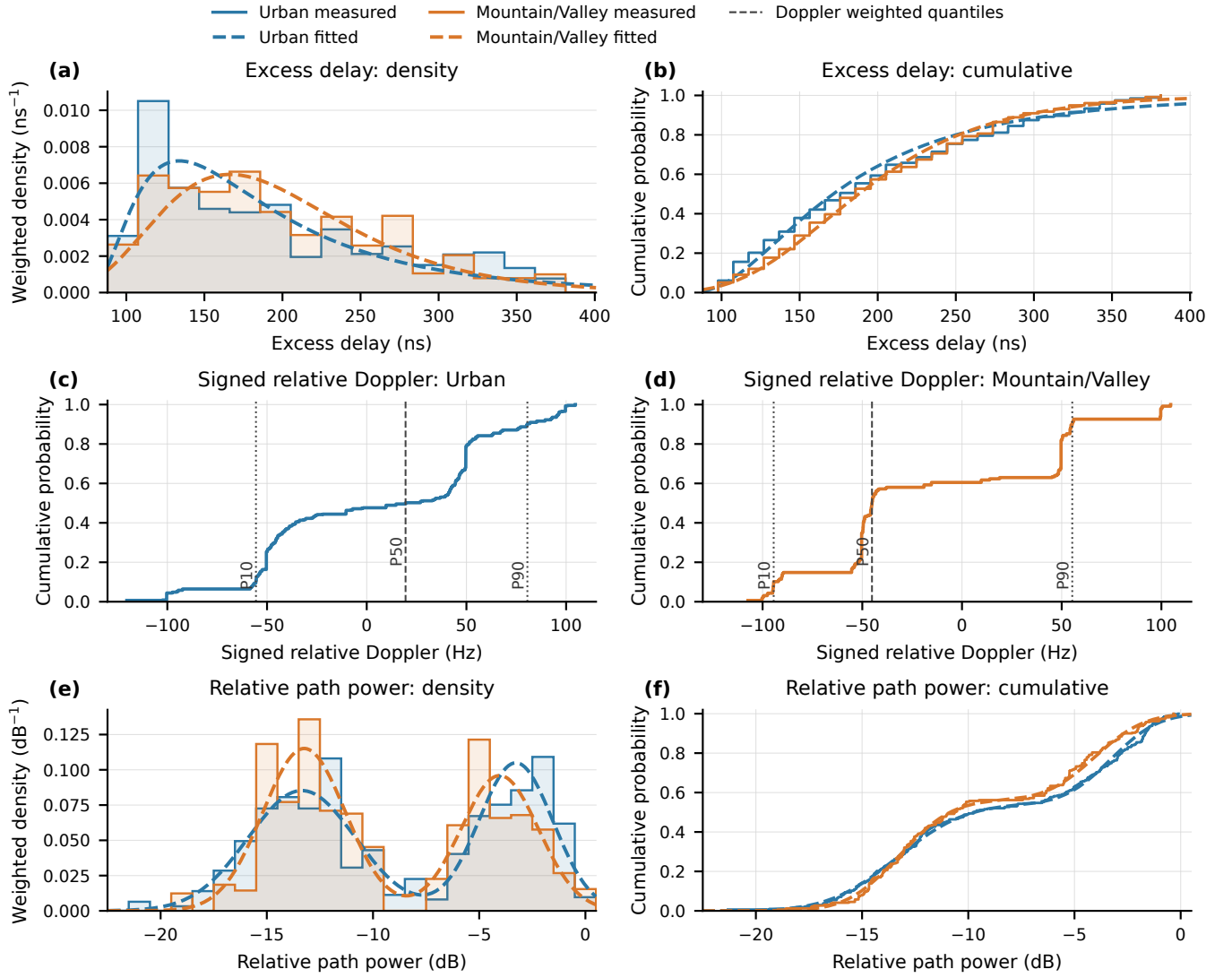


Fig. 2. Measured path-parameter distributions and statistical descriptions: (a)–(b) excess delay, (c)–(d) track-balanced empirical CDFs of signed relative Doppler for Urban and Mountain/Valley, respectively, and (e)–(f) relative path power. Fitted curves are shown for excess delay and relative power.

rates, temporal evolution, or a complete joint stochastic channel model.

TABLE I
PARAMETERS OF THE ENVIRONMENT-SPECIFIC DESCRIPTIONS.

Quantity	Environment	Model summary	Fit measure
Excess delay	Urban	Shifted lognormal: shift 68.7 ns, scale 102.8 ns, log-std. 0.675	$D_w = 0.082$
	Mountain/Valley	Lognormal: scale 187.8 ns, log-std. 0.348	$D_w = 0.055$
Signed relative Doppler	Urban	Track-balanced empirical CDF: $P_{10} = -55.6$, $P_{50} = 19.5$, $P_{90} = 80.6$ Hz	—
	Mountain/Valley	Track-balanced empirical CDF: $P_{10} = -94.4$, $P_{50} = -45.1$, $P_{90} = 55.3$ Hz	—
Relative power	Urban	GMM: $(-13.35, 2.53, 0.541)$ and $(-3.27, 1.74, 0.459)$ dB	$D_w = 0.045$
	Mountain/Valley	GMM: $(-13.24, 1.95, 0.562)$ and $(-4.03, 1.82, 0.438)$ dB	$D_w = 0.037$
Power-delay	Urban	Breakpoint 190.6 ns; slopes -3.53 and -5.57 dB/100 ns	RMSE 4.01 dB
	Mountain/Valley	Breakpoint 190.6 ns; slopes -6.31 and -4.46 dB/100 ns	RMSE 3.44 dB

For each GMM component, the three numbers are mean, standard deviation, and weight. D_w is the maximum weighted CDF deviation defined in (5) and applies to the fitted delay and relative-power marginals. Signed relative Doppler is summarized by weighted empirical quantiles.

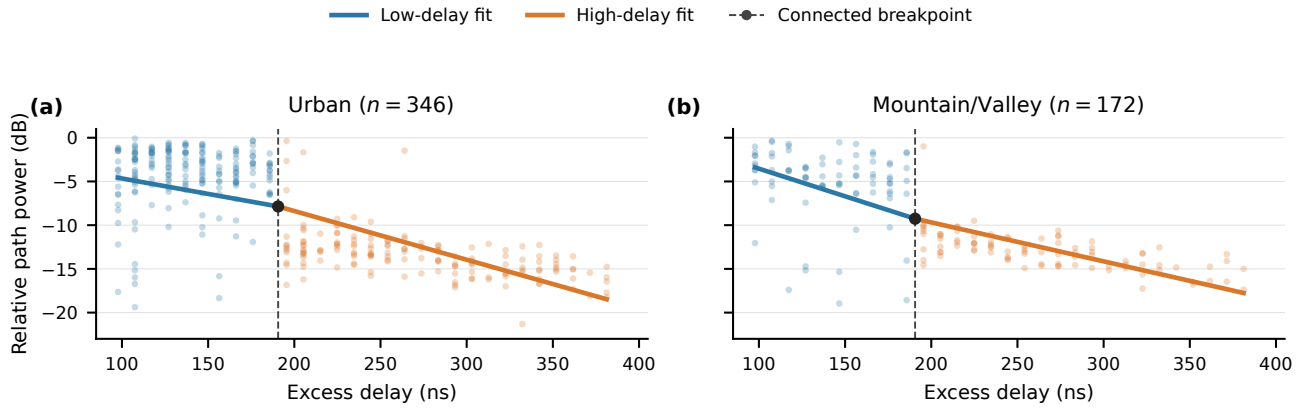


Fig. 3. Measured relative power and connected two-slope fits versus excess delay.

D. Power–Delay Relation

Fig. 3 shows the decrease in relative power with increasing excess delay. The connected model has a common breakpoint of 190.6 ns, with one fitted mean power-decline rate below and another above it. In Urban, the slopes are -3.53 and -5.57 dB per 100 ns before and after the breakpoint; in Mountain/Valley, they are -6.31 and -4.46 dB per 100 ns. The fitted mean trend is steeper after the breakpoint in Urban and less steep after it in Mountain/Valley. The root-mean-square errors are 4.01 and 3.44 dB, respectively.

V. CONCLUSION

This paper presents measurement-based extraction and statistical description of GPS L1 C/A MPCs from moving-vehicle measurements. Navigation-data alignment, per-satellite SAGE estimation, and temporal consistency provide persistent observations of excess delay, signed relative Doppler, and relative power. The resulting descriptions capture the right-skewed delay distribution, the measured signed relative Doppler distributions, the two main modes of relative power, and the change in mean power-decline rate between low-delay and high-delay MPCs. Unlike receiver-level multipath indicators, these descriptions operate directly on individual MPC parameters. The fitted delay and power distributions and the empirical CDF of signed relative Doppler provide measurement-based parameter inputs for simplified GNSS channel simulations and receiver evaluation in Urban and Mountain/Valley propagation scenarios.

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